

RELAY

Learning When and Whom to Communicate With in Heterogeneous Multi-Agent Search and Rescue

Research Design & Experimental Framework

Version 0.6 | 15 August 2026 | Internal project specification

Scout • Ambulance • Fireman

MAPPO / CTDE • Emergent WHEN / WHO / WHAT • Context-dependent routing

Document status: replaces the previous “Architecture & Experimental Design” naming. This document defines the research design, experimental logic, system requirements, evaluation protocol, and implementation boundaries; it is not the environment specification or hyperparameter sheet.

Document Map

This version follows a research-design flow: define the question first, then the system, experiments, validation, controls, and implementation requirements. The Objectives section contains direct links to the detailed experiment sections.

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Reading shortcut

For the project logic, read Sections 1-3 and 8-10. For implementation, read Sections 4-7 and 11-16. The full literature reviews remain in the separate RELAY Extended Literature Survey.

1. Project Definition & Research Scope

RELAY studies emergent communication in a cooperative, partially observable, heterogeneous multi-agent search-and-rescue task. The pilot uses three specialized agents - Scout, Ambulance, and Fireman - in a 2D environment containing victims and fires. Agents must learn useful task behaviour while also learning whether communication is worth its cost and which teammate should receive information.

Central research question

Can heterogeneous cooperative agents learn when communication is worth its cost and whom task-relevant information should be sent to, using only task reward rather than predefined message semantics?

1.1 Confirmed scope

- Three specialized roles: Scout, Ambulance, Fireman.

- Victims require Ambulance intervention; fires require Fireman intervention; Scout is the primary exploration/sensing specialist.
- Partial observability at execution: no actor receives the full world state.
- Centralized training with decentralized execution (CTDE) using recurrent MAPPO as the common learning backbone.
- Communication is learned and decomposed into separate WHEN, WHO, and WHAT decisions.
- Communication has an explicit per-recipient transmission-attempt cost.
- Primary research focus: WHEN, WHO, and the stronger context-dependent WHO extension.
- Validation focus: communication utility and causal/semantic analysis of WHAT.
- Optional goals only after the primary objectives are stable: latency/loss robustness, capacity, and discrete communication.
- Headless training and visual checkpoint replay use the same deterministic environment core.

1.2 Document boundary

This document deliberately does not finalize map dimensions, exact observation tensors, sensor radii, movement/rescue mechanics, reward magnitudes, PPO hyperparameters, or the concrete simulator package. Those belong in the Environment + Observation + Action + Reward Technical Design and implementation configuration. The research design must remain stable even if those engineering choices change.

2. Objectives

Objectives are research questions. Tunable quantities such as communication cost, message dimension, delay, sensor range, or PPO settings are parameters - not objectives.

2.1 Priority Questions

P1 - WHEN - Selective Communication

Can agents learn when transmitting information is useful enough to justify its communication cost?

- Learn SEND vs SILENCE from task reward rather than hard-coded events.
- Compare learned gating with always-on communication under otherwise matched conditions.
- Check whether communication concentrates around task-relevant events while preserving task performance.

[Go to §8.1 - P1: WHEN](#)

P2 - WHO - Selective Recipient Routing

Can agents learn which teammate should receive information instead of indiscriminately broadcasting it?

- Use hard sender-side routing; WHEN and WHO remain separate decisions.
- Baseline pilot uses Scout / Ambulance / Fireman.
- Keep claims conservative: the three-agent setting may partly reduce to event-type routing.

[Go to §8.2 - P2: WHO](#)

P3 - Context-Dependent WHO - Individual Teammate Selection

Can routing depend on the state of individual teammates rather than merely their role?

- Extend to one Scout, two Ambulances, and two Firemen after the three-agent system is stable.
- Specialists may learn to communicate their own status/state without supervised BUSY/FREE symbols.
- Scout uses received messages plus GRU memory to choose an actual specialist ID.
- Test whether routing changes with availability, workload, prior knowledge, and - where the environment supports it - distance/reachability.

[Go to §8.3 - P3: Context-Dependent WHO](#)

2.2 Validation & Protocol Analysis

V1 - Communication Utility

Does communication actually improve the cooperative task?

- NoComm vs always-on communication is a prerequisite sanity check.
- If communication is not useful, the environment must be redesigned before P1-P3 claims are meaningful.

[Go to §9.1 - Communication Utility](#)

V2 - WHAT - Learned Message Function

Did the learned payload acquire causally useful task information?

- Use force-silence, message scrambling, and recipient interventions on frozen policies.
- Probe relationships between messages and discoveries, specialist state, task events, and spatial/task context.
- Do not assign human semantics to a latent message unless interventions support the interpretation.

[Go to §9.2 - WHAT / Causal Analysis](#)

2.3 Optional Goals

O1 - Channel Robustness

How much does a clean-trained protocol degrade under basic latency and packet loss?

- Fixed latency and independent packet loss only in the primary optional study.
- Keep clean-trained zero-shot testing separate from any future robustness-aware retraining.

[Go to §10.1 - Robustness](#)

O2 - Message Capacity

How sensitive is the protocol to smaller or larger continuous message representations?

- Treat latent dimension as a representation constraint, not as physical bit-rate.
- Only run after P1-P3 are stable.

[Go to §10.2 - Capacity](#)

O3 - Discrete Communication

Can useful coordination be retained with a finite categorical/symbolic message vocabulary?

- Compare against the continuous baseline while keeping the MAPPO backbone and task conditions matched.
- Do not claim N-bit communication without explicit symbol/entropy accounting.

[Go to §10.3 - Discrete Communication](#)

3. Research Gap, Hypotheses & Expected Contributions

3.1 Research gap motivating RELAY

The surveyed communication-MARL literature contains strong treatments of selective timing, recipient targeting, heterogeneous teaming, bandwidth constraints, receiver-side information integration, and channel impairment. However, these questions are often isolated or bundled inside large specialized architectures. RELAY intentionally keeps a common MAPPO/CTDE backbone and asks a narrower, testable question: whether heterogeneous agents can learn selective communication and individual teammate routing in a rescue setting without predefined message semantics.

- Basic role-targeted WHO can be too easy when there is only one Ambulance and one Fireman; the multi-specialist experiment is designed to test context-dependent selection rather than event-type classification.
- Communication efficiency has at least two distinct parts: how often agents transmit and how much information a transmission can carry. RELAY keeps the first part central and treats capacity/discreteness as optional follow-up.
- Low-dimensional continuous messages must not be mislabeled as physical bandwidth or bits.
- Latency and packet loss create different failure modes and should be evaluated separately rather than mixed into one vague “robustness” result.
- Protocol meaning must be supported by causal interventions and probing, not inferred from reward improvement alone.

3.2 Working hypotheses

- H1 - Selective communication can retain useful cooperative performance while reducing unnecessary transmissions relative to always-on communication.
- H2 - Heterogeneous agents can learn task-relevant sender-side recipient routing without predefined recipient rules.
- H3 - With multiple same-role specialists, emergent status communication plus recurrent memory can support context-dependent individual routing.
- H4 - Learned messages will show measurable causal relevance: removing, scrambling, or misrouting them should degrade the behaviours that depend on them.

3.3 Expected contribution

- A reproducible heterogeneous rescue testbed for emergent communication under partial observability.
- A clean WHEN / WHO / WHAT decomposition with sender-side routing and a common MAPPO training backbone.
- A context-dependent multi-specialist routing experiment that tests whether learned communication supports dynamic teammate selection.
- Causal and behavioural evidence for what the learned protocol is doing, rather than relying only on return curves.
- Optional extension results on basic channel impairment, representation capacity, or discrete communication if the primary study is completed.

4. RELAY System Overview

The architecture is modular because the research variable is communication. World dynamics, actor observations, learning backbone, and logging should remain stable while communication mechanisms or channel settings are changed.

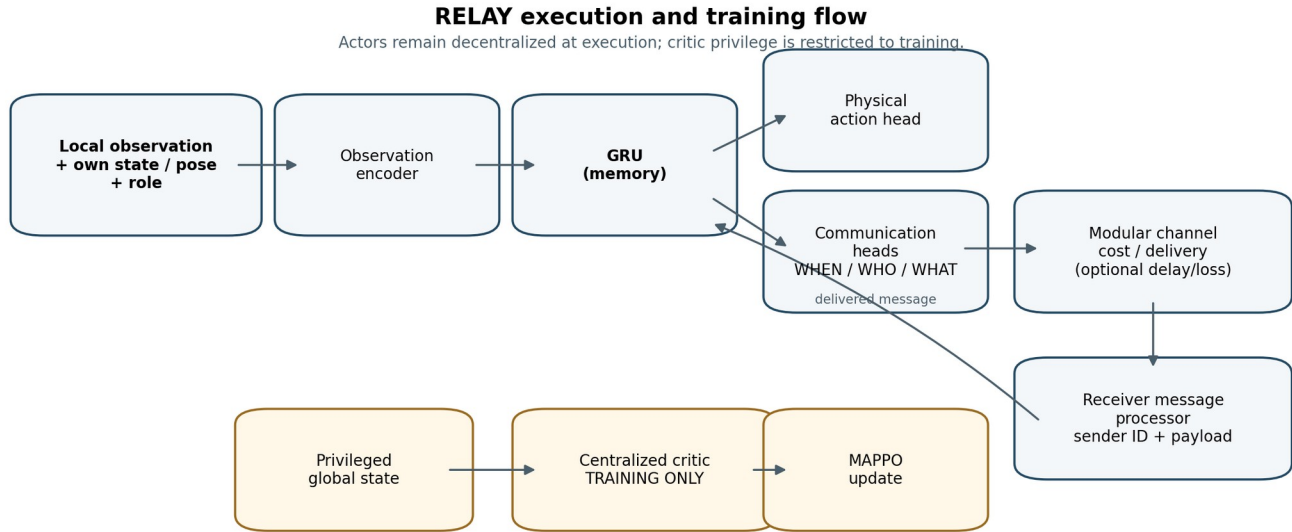


Figure 1. RELAY execution and training flow. Communication is an explicit subsystem between sender policy output and receiver policy input.

4.1 Execution loop

1. Environment produces each agent's local observation and maintains the true world state.
2. Actor processes execution-legal information only: local observation, own state/pose, role, recurrent state, and messages delivered this timestep.
3. Actor produces a physical action and communication decisions.
4. Communication subsystem applies WHEN/WHO/WHAT, cost accounting, delivery, and any optional channel impairment.
5. Receiver message processor converts sender metadata plus payload into features that enter the recurrent policy.
6. GRU carries forward temporal information; there is no separate ever-growing raw message-history transcript.
7. Visual mode reads the same environment/log stream and never changes policy inputs or simulation timing.

4.2 Training loop - CTDE

MAPPO uses decentralized recurrent actors and a centralized critic. The critic may use privileged compact global state and useful agent-specific features during training, but this information is never exposed to deployed actors. Keeping the learner fixed across communication variants is essential for attribution: communication should be the changed variable, not the RL algorithm.

5. Agent Roles, Information & Task Structure

Role	Primary capability	Key limitation	Communication relevance
Scout	Fast/strong exploration and sensing; discovers victims and fires efficiently.	Cannot rescue victims or extinguish fires.	Produces discoveries and may also receive specialist status information before routing tasks.
Ambulance	Victim-response specialist with local target sensing.	Cannot extinguish fires; sensing weaker than Scout.	Needs useful victim/location information; in P3 may communicate its own status/state.
Fireman	Fire-response specialist with local hazard sensing.	Cannot rescue victims; sensing weaker than Scout.	Needs useful fire/location information; in P3 may communicate its own status/state.

5.1 Information boundary

- Each agent knows its own state and pose in a common coordinate frame.
- Each agent observes only locally visible/sensed world information.
- Remote target locations, teammate states, workload, and remote teammate position are not automatically exposed unless the chosen environment mechanic makes them locally observable.
- Remote knowledge must therefore come from local sensing, memory, or delivered communication.
- Sender identity and sender role travel as packet metadata; this is treated like network addressing information, not hidden task-state leakage.

5.2 Why specialists retain local sensing

Ambulance and Fireman must not be completely blind to their own task objects. If a specialist can act only after the Scout communicates, NoComm becomes nearly impossible and communication utility becomes trivial. Limited local sensing keeps independent search feasible while allowing communication to improve discovery-to-response time, route efficiency, and completion.

5.3 Multi-specialist configuration for P3

P3 expands the team to one Scout, two Ambulances, and two Firemen. WHO then selects individual agent IDs rather than only roles. The experiment is deliberately staged after the three-agent pilot because five-agent credit assignment and communication history are harder. Specialists are not given supervised “busy/free” message labels; any useful status code must emerge because it improves shared task return.

6. Agent Policy Design

6.1 Inputs and observation encoding

The actor receives local observation, own pose/state, role identity, messages delivered this timestep, and its recurrent hidden state. An observation encoder is simply the neural module that converts the raw local observation into compact features suitable for the policy. Its exact form is deferred until the observation representation is fixed.

6.2 GRU memory

Confirmed design

The GRU is the message-history mechanism. RELAY does not feed an ever-growing raw conversation transcript to the actor. Previous observations and delivered messages influence future decisions through recurrent state.

6.3 Shared vs role-specific processing

The starting point remains a shared/general representation with role conditioning and role-specific physical-action outputs. The heterogeneous-team literature supports allowing asymmetric role behaviour, but RELAY will not import a full heterogeneous graph architecture into the baseline. Fully separate policies or role-pair-specific communication modules become troubleshooting/future ablations only if shared processing causes clear interference.

6.4 Output heads

- Physical action head - role-appropriate environment action distribution.
- WHEN head - SEND or SILENCE.
- WHO head - intended recipient(s), evaluated separately from WHEN.
- WHAT head - small learned message payload with no predefined semantics.

7. Communication Design

7.1 Packet structure

A delivered packet conceptually contains sender identity/role, intended receiver, learned payload, and channel bookkeeping needed by the environment/logger. The payload itself does not contain supervised semantic labels.

7.2 WHEN - transmit or remain silent

WHEN is a learned SEND/SILENCE decision. No rule such as “send whenever a victim is detected” is hard-coded. A communication cost creates a real trade-off: information must improve expected team outcome enough to justify transmission.

7.3 WHO - sender-side routing

WHEN and WHO remain distinct. If WHEN = SEND, WHO chooses the intended recipient set. In the three-agent pilot, WHO routes between the two teammates. In P3, WHO chooses individual agents such as A1, A2, F1, and F2. This is sender-side routing: information is not broadcast to everyone and filtered only after delivery.

7.4 WHAT - learned payload

The baseline uses a small fixed continuous message vector. The exact initial dimensionality is an implementation parameter to be fixed before training. Message meaning is learned from task reward and diagnosed after training. A d-dimensional float vector is a representation constraint; it must not be described as a d-bit channel.

7.5 Receiver message processing

For each delivered packet, the receiver combines sender identity/role with the payload and passes them through a small neural message processor. If several packets arrive in the same timestep, a simple deterministic combination rule (for example fixed-order concatenation with masks, or masked sum/mean after per-packet processing) produces the communication feature given to the GRU. Attention/Transformer machinery is intentionally excluded unless the simple mechanism proves inadequate.

7.6 Communication learning semantics - OPEN DECISION

Decision required before coding the communication learner

Two training semantics remain possible: (A) treat WHEN / WHO / WHAT as MAPPO policy outputs/actions, preserving a clean channel boundary and making discrete/lossy extensions natural; or (B) allow a differentiable continuous sender-to-receiver pathway for WHAT, which may learn faster but changes the training semantics. The project should select one as the primary mechanism before implementation rather than mixing both accidentally.

7.7 Communication cost

Cost is charged per sender-recipient transmission attempt. Silence costs zero; a send to one recipient costs `communication_cost`; sending to two recipients costs twice that amount. The attempt is charged even if an optional impaired channel later drops or delays the packet. `communication_cost` is a tunable experimental parameter, not an objective and not a scientific constant.

8. Priority Experiment Design

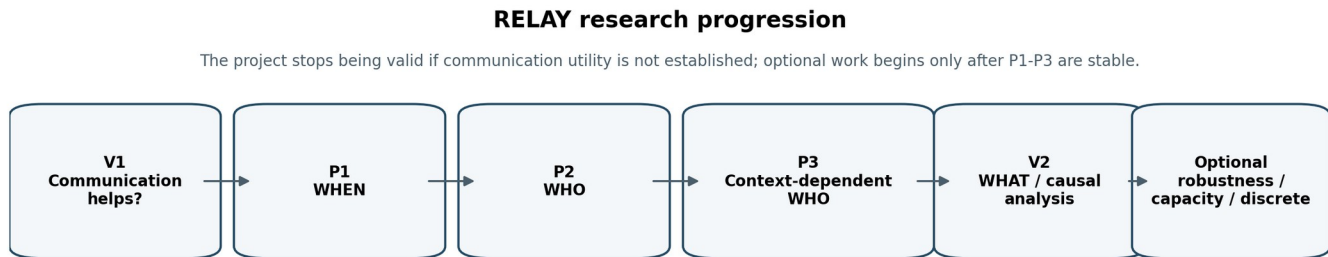


Figure 2. Research progression. V1 is a validity gate; P1-P3 form the core study; V2 establishes causal evidence; optional goals come last.

8.1 P1 - WHEN: Selective Communication

Question: Can agents learn to remain silent when communication is not worth its cost?

Core comparison: Always-on broadcast vs WHEN-only, with the same environment, message representation, MAPPO backbone, and training budget.

Mechanism: WHEN-only learns SEND/SILENCE; if SEND, the message is broadcast to all eligible teammates so WHO is not part of this comparison.

Parameter study: After the mechanism is stable, evaluate a small low/medium/high communication-cost sensitivity sweep; this is a parameter sweep, not a separate objective.

Evidence required: Task performance remains competitive while communication rate/cost falls, and communication events concentrate around useful task changes rather than every timestep.

8.2 P2 - WHO: Selective Recipient Routing

Question: Can agents learn to route task-relevant information to the appropriate teammate rather than broadcast?

Core comparison: Always-on broadcast, WHO-only (always send, learned recipient), and full WHEN+WHO.

Pilot team: One Scout, one Ambulance, one Fireman.

Guardrail: With only two specialist recipients, WHO may partly reduce to event-type classification. Claims at this stage should therefore be “learned task-relevant routing,” not sophisticated context-dependent allocation.

Evidence required: Selective routing reduces wasted recipient deliveries and/or improves coordination without relying on predefined victim-to-Ambulance or fire-to-Fireman rules.

8.3 P3 - Context-Dependent Multi-Specialist WHO

Question: Can the communication protocol support selection among multiple valid same-role teammates based on current context?

Team: One Scout, two Ambulances, two Firemen.

Key mechanism: Specialists can communicate their own state through the same learned channel. Scout receives those messages, retains useful history through the GRU, and WHO chooses an individual agent ID.

No semantic supervision: The environment never labels a payload as BUSY, FREE, CLOSE, or AVAILABLE. Any such functional meaning must be inferred from task reward and later analysis.

Controlled contexts: A1 busy/A2 free; A2 busy/A1 free; both free; both occupied; one specialist already informed; and, if represented cleanly by the environment, distance/reachability differences.

Evidence required: Recipient choice changes systematically with specialist context, and disabling/scrambling specialist-to-Scout status communication weakens that context sensitivity.

Why P3 matters

P3 addresses the main weakness of the three-agent WHO claim. In the pilot, victim vs fire can predict the recipient role. In P3, two Ambulances share the same role, so successful routing must depend on individual context rather than object class alone.

9. Validation & Protocol Analysis

9.1 V1 - Communication Utility

Before interpreting emergent communication, verify that the task actually benefits from information exchange. Compare a strong NoComm MAPPO baseline with always-on communication using identical observations, reward, model capacity (apart from the communication path), optimizer family, rollout budget, and evaluation protocol. If always-on communication does not provide a meaningful advantage, redesign task information dependencies before running P1-P3.

9.2 V2 - WHAT / Causal Message Analysis

Force silence: Freeze a trained model and block message delivery. A drop in the behaviours that rely on communication supports causal utility.

Message scrambling: Preserve approximately the same communication rate but deliver payloads from wrong timesteps/episodes. A degradation indicates that content, not mere communication occurrence, matters.

Recipient intervention: Redirect or randomize recipients during evaluation. A drop indicates that learned routing and receiver-specific use of information matter.

Event association: Probe whether message patterns correlate with victim/fire discovery, specialist state, task completion, spatial context, or other observable events.

Temporal redundancy: Measure whether nearly unchanged messages are repeatedly sent over adjacent timesteps. This is analysis only; RELAY does not add a TMC-style temporal controller to the core model.

Message usage caution: Do not label a latent value as “victim,” “busy,” or similar solely because a cluster appears correlated. Use intervention evidence where possible.

9.3 Contextual WHO analysis

P3 needs more than total return. Report conditional routing statistics: for example, when A1 is occupied and A2 is available, how often is a new victim routed to A2? Then reverse the condition. Repeat with specialist-status communication disabled or scrambled. This directly tests whether WHO is using learned teammate context.

10. Optional Research Goals

Scope rule

Sections 10.1-10.3 are attempted only after P1-P3 and their validation analyses are stable across multiple seeds. They are not required for completion of the primary study.

10.1 O1 - Basic Channel Robustness

Primary optional protocol: train under ideal delivery, freeze the checkpoint, then evaluate separately under fixed latency and independent packet loss. Do not combine clean-trained zero-shot testing with robustness-aware retraining under one label. More elaborate variable delay, burst loss, message-age models, or channel-aware retraining are outside the committed scope.

10.2 O2 - Message Capacity

Run a small continuous message-dimension sensitivity study only if the core model is stable. Report it as a representation-capacity experiment. Do not convert dimensions to “bits” or “bandwidth” without explicit quantization/entropy accounting.

10.3 O3 - Discrete Communication

Replace the continuous payload with a finite categorical vocabulary while keeping the task and MAPPO backbone matched as closely as practical. This is the clean path toward a symbolic protocol. True bit-rate claims require a defined symbol alphabet/coding rule and are beyond the baseline continuous experiment.

11. Experimental Controls & Parameters

11.1 Fair-comparison rule

Across communication ablations, reuse the same environment distribution, observation interface, reward definition, actor/critic family, optimizer family, rollout budget, random-seed protocol, and evaluation procedure wherever possible. Only the communication mechanism or explicitly tested parameter should change.

11.2 Parameter classes

Class	Examples
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Communication	communication_cost, message dimension, maximum recipients, optional delay, optional packet-loss probability
Environment	map/generator, object counts, sensor ranges, movement/rescue mechanics - owned by Environment Spec
Training	learning rate, PPO clip, rollout length, epochs, batch size, GRU size, seeds - owned by training config

11.3 Communication-cost sensitivity

Communication cost is a parameter sweep attached to the final selective-communication model. Use a small set of low/medium/high values chosen relative to the finalized reward scale. The purpose is to plot the task-performance vs communication-usage trade-off, not to create a new objective.

12. Evaluation Metrics & Success Criteria

Metric family	Examples / evidence
Task performance	success rate; episode return; victims resolved; fires resolved; completion steps/time
Communication efficiency	transmission attempts; delivered recipient count; silence rate; communication cost; messages by role
Routing quality	recipient distribution by event/context; unnecessary recipient deliveries; broadcast vs targeted cost
Coordination	discovery-to-response time; redundant search; distance/steps to intervention; task allocation efficiency
P3 contextual routing	conditional recipient choice under busy/free, informed/uninformed, and other controlled specialist states
Learning	convergence speed; sample efficiency; variance across seeds; failure/local-minimum rate
Optional robustness	performance and communication degradation vs delay/loss level
Protocol evidence	force-silence/scramble sensitivity; event-message correlations; routing intervention sensitivity

12.1 Minimum success criteria for claims

- Communication utility: Always-on communication must outperform or meaningfully improve coordination relative to NoComm on held-out evaluation.
- P1: learned WHEN must show a useful task-performance / communication-usage trade-off rather than merely collapsing to always-send or never-send.

- P2: learned routing must reduce irrelevant recipient deliveries and/or improve coordination relative to broadcast under matched conditions.
- P3: recipient choice must change with individual specialist context, and this dependence must weaken when relevant status communication is disrupted.
- WHAT: at least one causal intervention must demonstrate that learned payload content contributes beyond the occurrence of communication itself.
- Final claims require multiple random seeds and held-out map/environment randomizations; checkpoint-selection rules must be defined before final reporting.

13. Validity Guardrails & Failure Conditions

Non-trivial communication: NoComm must remain possible but meaningfully less efficient than useful communication. If specialists are fully dependent on Scout messages, utility becomes trivial; if every agent sees everything, communication becomes unnecessary.

No hard-coded protocol: Rewards/observations may make communication useful, but must not label token meanings, recipient rules, or BUSY/FREE semantics.

Clean isolation: Do not simultaneously change sensing, reward scale, channel cost, and task difficulty when claiming an effect from one communication mechanism.

Conservative WHO claims: Three-agent P2 establishes task-relevant routing. Strong context-dependent selection claims are reserved for P3.

Strong baseline: Communication gains must be measured against a competent MAPPO NoComm baseline, not against a deliberately weak learner.

Interpretability discipline: Attention plots or message clusters are descriptive. Causal message removal/scrambling/routing interventions are stronger evidence.

Stop conditions: If the base three-agent task does not learn reliably across seeds, do not add five-agent routing, latency, discrete communication, or other complexity. Fix the environment/training first.

14. Logging, Replay & Reproducibility

Logging is a first-class research requirement because reward curves cannot establish WHEN/WHO behaviour or message meaning. The system should preserve enough information to reproduce both quantitative results and visually interesting success/failure episodes.

14.1 Always store per run

- Git commit/code version and experiment config
- random seed and environment-generator/config identifiers
- algorithm/model architecture and communication mode
- training hyperparameters and checkpoint IDs
- compact episode summaries and training diagnostics

14.2 Evaluation trajectory log

- world/event timestep, discoveries, interventions, agent positions and task state
- physical action; WHEN output; WHO intended recipients
- WHAT payload or token; sender identity/role
- attempted transmissions, delivered recipients, communication cost
- optional queue delay/drop outcome when robustness is enabled
- receiver-delivered messages and subsequent action/state change

14.3 Training diagnostics and storage tiers

Track PPO policy/value loss, entropy, approximate KL, clip fraction, explained variance/value statistics, gradient norms, and rollout/update identifiers. Store full trajectories for all evaluation episodes and selected training checkpoints; store heavy internals such as full GRU states, logits, or embeddings only for selected diagnostic episodes. Columnar formats such as Parquet suit timestep/event logs; YAML/JSON suit configs and small metadata.

15. Execution & Environment Interface

15.1 One environment core, two modes

- Headless mode: fast stepping, parallel/vectorized rollouts where practical, deterministic seed control.
- Visual replay mode: checkpoint playback and diagnostics over the same state/log stream; rendering must never alter dynamics or observations.

15.2 Minimum visual diagnostics

- agent positions/roles and physical paths/actions
- local sensing/visible regions
- victims, fires, intervention state, reward and mission progress
- SEND vs SILENCE and sender-to-recipient routing lines
- attempted vs delivered communication counts and communication cost
- message/token identifier or compact continuous-message summary
- optional delay queue and drop events only when those experiments are enabled

15.3 Environment contract

- Actor interface exposes execution-local observations only.
- Critic interface exposes privileged training state separately.
- Channel hooks permit cost accounting, routing, delivery, and optional impairment without changing world dynamics.
- Environment is deterministic/seedable and replayable for diagnostics.
- Environment framework must support rescue-specific partial observations, role capabilities, logging, and vectorized/headless execution without excessive adaptation cost.

16. Implementation Stack & Engineering Posture

Layer	Current plan
Language / learning	Python, PyTorch
MARL training	Recurrent MAPPO with centralized critic
Environment core	Lightweight existing multi-agent substrate where practical; custom rescue task mechanics
API	Gymnasium / PettingZoo-compatible interface preferred where useful
Execution	Headless/vectorized training plus deterministic visual replay
Configuration	YAML/Hydra-style experiment configuration
Logging	Weights & Biases or TensorBoard for curves; Parquet/JSON/YAML for structured logs

Analysis	NumPy, Pandas, Matplotlib; scikit-learn for simple probing/PCA if needed
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16.1 Reuse substrate, customize task

Do not rebuild stepping, batching, reset/seeding, API plumbing, or rendering infrastructure by default. Compare a thin custom rescue task on an existing substrate against a purpose-built lightweight gridworld. Use the existing substrate if it saves engineering work without constraining partial observability, role mechanics, communication hooks, determinism, or training throughput.

17. Scope & Decision Register

17.1 Confirmed

- RELAY title/tagline and research focus on WHEN, WHO, and context-dependent WHO.
- Three-role pilot with separate five-agent multi-specialist P3 extension.
- MAPPO/CTDE with recurrent decentralized actors.
- GRU as temporal/message-history memory.
- Separate WHEN and WHO decisions.
- Hard sender-side routing.
- Learned WHAT semantics; small fixed continuous baseline payload.
- Sender identity/role included as packet metadata.
- Simple receiver-side message processor before recurrent integration.
- Per-recipient transmission-attempt cost.
- Modular communication channel and full communication logging.
- NoComm / always-on validity baseline and causal WHAT analysis.

17.2 Optional only

- fixed latency / independent packet-loss robustness
- message-capacity sensitivity
- continuous vs discrete communication

17.3 Explicitly outside current committed scope

- multi-round communication
- range-restricted communication
- dynamic/variable message sizing
- full HetNet graph architecture
- M2I2-style intention inference / masked-state reconstruction
- CoDe/VIL2C specialized delay architectures
- TMC temporal-controller implementation
- Engineer/fourth-role extension
- TarMAC receiver-attention comparator unless later needed for a specific paper comparison

17.4 Open decisions before implementation

- Primary WHAT training semantics: MAPPO communication action vs differentiable continuous channel.
- Initial continuous message dimension.
- Exact receiver multi-message combination rule.
- Concrete environment substrate and observation representation (resolved in the Environment Technical Design).

18. Literature Basis & Design Trace

The literature informs experimental design and terminology; RELAY is not a reproduction of any single paper. Detailed reviews are maintained separately in the RELAY Extended Literature Survey.

Source family	Design implication retained in RELAY
Foerster et al. / RIAL-DIAL	Emergent communication, recurrent agents, discrete vs differentiable communication, causal message ablations.
CommNet	Continuous communication baseline and NoComm/broadcast comparison logic.
IC3Net + ETCNet + sparse communication work	Separate WHEN gate; communication must have a real downside/cost; analyze communication rate around meaningful events.
TarMAC + T2MAC	Recipient relevance and targeted communication; distinguish hard sender routing from receiver-side attention.
MAPPO	Strong CTDE/PPO backbone and controlled comparison discipline.
HetNet	Heterogeneous roles can justify asymmetric communication; do not force every role through identical semantics. Full graph architecture remains unnecessary for baseline.
IMAC / BVME / Effective Limited-Bandwidth / DSMS	Capacity and information-content constraints are distinct from transmission frequency; latent dimension is not physical bits.
DRMAC / M2I2 / COCOM	Receiver integration, redundancy, and receiver knowledge matter; use these ideas mainly for analysis/interpretation rather than adding complex modules.
CoDe / VIL2C / TMC	Delayed, stale, and lossy communication create separate robustness questions; simple fixed latency/loss is sufficient for optional RELAY scope.
Freed et al. discrete noisy channel	Discrete execution is feasible but requires deliberate training/interface choices; motivates optional discrete communication.

18.1 Terminology rules carried forward

- WHEN = whether to transmit; WHO = intended recipient; WHAT = learned payload/content.
- Hard sender routing is not equivalent to broadcasting and later applying receiver attention.
- Continuous vector dimension, discrete vocabulary size, entropy, and physical bit-rate are different quantities.
- Clean-trained zero-shot channel stress testing and impairment-aware retraining answer different questions.
- Interpretability claims require empirical intervention/probing support.

19. Deliverables

- RELAY environment/interface satisfying the actor, critic, channel, logging, seeding, headless, and replay requirements.
- Trained three-agent baseline and WHEN/WHO variants with multi-seed evaluation.
- Context-dependent multi-specialist P3 experiment if the three-agent system is stable.
- Communication utility and causal WHAT validation analyses.

- Reproducible checkpoints, configs, structured logs, and visual replay demonstrations.
- Final research report/paper with controlled comparisons and conservative claims.
- Optional robustness, capacity, and/or discrete communication results only if primary objectives are complete.

Next design document

Environment + Observation + Action + Reward Technical Design. It should resolve the concrete rescue mechanics, map generation/seed validation, sensor model, observation tensors, action semantics, reward scale, and initial difficulty while preserving the research boundaries defined here.

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END OF RELAY RESEARCH DESIGN & EXPERIMENTAL FRAMEWORK v0.6